

NovaMart Intelligence Platform

Actual Application UI Frame Screenshots Captured Live from Production

 Verified Production Interface Screenshots (10 Frames)

This document contains unedited 1440x900 screenshots captured directly from the live Vercel production deployment at <https://kpi-engine.vercel.app> using automated Puppeteer headless navigation. It documents all 9 main application screens plus the slide-in Analyst Chat drawer, confirming live functionality across movement detection, LMDI factor splits, abstention protocols, live time simulation, RBAC redactions, and telemetry monitoring.

 Table of Live UI Frames Included:

1. **Overview Dashboard** — KPI Cards, LMDI Bar Chart, Confidence Gauge

2. **Insight Analysis** — Persona Narrative, What-If Sandbox Sliders, Lineage

3. **Abstention Protocol** — Low Confidence Alert, Failure Logs, Clarification Chips

4. **Driver Attribution** — OLS Panel Regression Table, Pearson/Spearman Matrix

5. **Live Simulator Lab** — Parameter Shocks, Sprint Trigger, SQLite Counters
6. **Data Sources & Ingestion** — SLA Freshness, Row Counts, JSON Editor

7. **Semantic Layer** — KPI Contracts, Formulas, Owners, Materiality

8. **Feedback Loop** — Analyst Helpfulness Rating, Error Categorization

9. **System Health** — Latency Bar Chart, P95 Telemetry, Cost Routing

10. **Analyst Chat Drawer** — Natural Language Queries, Intent Metadata Badges

1. Overview Dashboard (Happy Path)

Revenue Decline Scenario — Prioritized KPI Cards, Materiality Scores & 5-Signal Confidence Gauge

ROUTE: / (OVERVIEW)

1. Overview Dashboard (Happy Path)

Live UI Components & State

- Top bar shows active Scenario dropdown (Revenue Decline) and Role selector (Executive).
- KPI Cards: Revenue (\$18.4M, -14.4%, HIGH Priority), Orders, AOV, Gross Margin, Conversion Rate.

Feature Functionality & Engine Mechanics

- Primary Insight Card displaying computed business impact and Top Drivers attribution bar chart.
- Materiality Ledger ranking movements and 5-Factor Weighted Confidence Meter at 84% (HIGH).

2. Insight Analysis & What-If Sandbox

Validated Persona Narrative, LMDI Decomposition, Decision Sandbox & Traceable Lineage Ledger

ROUTE: / (INSIGHT TAB)

2. Insight Analysis & What-If Sandbox

🔍 Live UI Components & State

- Headline and Persona narrative with green "Output validated" pass badge and anti-causal guardrails.
- LMDI Factor Breakdown: Orders Volume Effect (-\$2.4M) vs AOV Price/Mix Effect (-\$694K).

⚡ Feature Functionality & Engine Mechanics

- Interactive "What-If" Decision Sandbox sliders for real-time projected dollar recovery.
- Traceable Evidence Ledger mapping claims to evidence IDs [E101-E110] with full lineage.

3. Selective Abstention Protocol (Contradictory Scenario)

Honest AI Refusal — Machine-Enforced Pause on Stale Data & Low Confidence

SCENARIO: CONTRADICTION EVIDENCE

3. Selective Abstention Protocol (Contradictory Scenario)

🔍 Live UI Components & State

- Amber warning panel replaces standard recommendations when data quality is insufficient.
- Explicit failure diagnostic reasons: Marketing data 29 hours stale, 18% missing external rows.

⚡ Feature Functionality & Engine Mechanics

- Prevents hallucinated advice when cross-source agreement drops below 50%.
- Interactive clarification prompt chips allowing analysts to query failure details directly.

4. Driver Attribution & Statistical Diagnostics

Multivariate OLS Panel Regression, Correlation Matrix & Anomaly Detection

ROUTE: / (DRIVERS TAB)

4. Driver Attribution & Statistical Diagnostics

🔍 Live UI Components & State

- Methodology banner: "Regression is associational, not causal proof."
- OLS Regression table displaying standardized beta strength bars, R^2 fit, and p-value significance.

⚡ Feature Functionality & Engine Mechanics

- Comparative Pearson r vs Spearman rank ρ correlation matrix for monotonicity checking.
- Auditable Business Rules triggers (SUPPLY_CONSTRAINT) and IsolationForest anomaly badges.

5. Live Time-Machine Scenario Simulator Lab

Parameter Shocks, Sprint Progression & Live SQLite Record Counters

ROUTE: / (SIMULATOR TAB)

5. Live Time-Machine Scenario Simulator Lab

🔍 Live UI Components & State

- Interactive parameter shock sliders: Marketing Spend (0.1x to 3.0x), Stockouts (0-50%), Price Index.
- Sprint Duration selector: Advance scenario timeline by 1, 7, or 14 days.

⚡ Feature Functionality & Engine Mechanics

- Live SQLite record counters tracking active transaction, marketing, and external row additions.
- One-click Scenario Reset button to restore pristine 365-day seeded benchmark.

6. Data Sources & Ingestion Portal

Pipeline SLA Freshness Audit, Schema Definitions & Custom JSON Ingestion

ROUTE: / (SOURCES TAB)

6. Data Sources & Ingestion Portal

🔍 Live UI Components & State

- Data freshness matrix auditing latency against expected SLAs for transactions, marketing, external feeds.
- Accordion inspection view showing grain, total rows, expected rows, and missing percentage.

⚡ Feature Functionality & Engine Mechanics

- One-Click "Seed All 4 Scenarios to Database" button powering instant PRNG re-seeding.
- Custom Data Ingestion Portal (POST /api/ingest) with JSON text area editor.

7. Semantic Layer & Metric Contracts

Central KPI Governance — Definitions, Formulas, Owners, Grains & Access

ROUTE: / (SEMANTIC TAB)

7. Semantic Layer & Metric Contracts

Live UI Components & State

- Centralized contract definitions for Revenue, Orders, AOV, Gross Margin, CAC, Stockout Rate.
- Formal mathematical formulas (e.g. $\text{SUM}(\text{unit_price} * \text{quantity})$, $\text{revenue} / \text{orders}$).

Feature Functionality & Engine Mechanics

- Business metric owners (VP Sales, Head of Growth, Supply Chain Lead).
- Associated statistical drivers, refresh cadences, materiality thresholds, and role access.

8. Feedback Loop & Analyst Telemetry

Human-in-the-Loop Analyst Ratings, Closed-Loop Weight Tuning & Feedback Audit

ROUTE: / (FEEDBACK TAB)

8. Feedback Loop & Analyst Telemetry

🔍 Live UI Components & State

- Analyst rating card: Helpful vs Not Helpful, category tagging, driver correction input.
- Feedback telemetry cards: Total insights reviewed, helpfulness percentage, error categories.

⚡ Feature Functionality & Engine Mechanics

- Explains closed-loop tuning: Analyst corrections dynamically penalize disproven driver weights.
- Audit log storing analyst corrections directly in PostgreSQL / SQLite.

9. System Health & Telemetry Dashboard

Latency Bar Chart per Request, P95 Metric, Cache Hit Rate & Cost Routing

ROUTE: / (HEALTH TAB)

9. System Health & Telemetry Dashboard

🔍 Live UI Components & State

- Telemetry metrics: Last latency (ms), P95 latency (ms), LLM calls avoided %, Cache hit rate %.
- Interactive Recharts bar chart tracking request latency across recent engine queries.

⚡ Feature Functionality & Engine Mechanics

- Cost-Optimized Model Routing breakdown: Tier 1 (Deterministic), Tier 2 (Fast), Tier 3 (Complex).
- Confirms 90%+ of quantitative queries skip LLM API token costs entirely.

10. Analyst Natural Language Chat Drawer

Natural Language Querying, Intent Classification & Cost-Tier Routing Badges

COMPONENT: ANALYST CHAT DRAWER

10. Analyst Natural Language Chat Drawer

Live UI Components & State

- Slide-in right drawer for conversational analytics in natural language.
- Assistant response pills displaying intent (why_down), filter tags (region=North), and execution latency.

Feature Functionality & Engine Mechanics

- Route indicator badge distinguishing narrative vs zero-token deterministic queries.
- Quick suggestion prompt chips for instant click-to-query execution.